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Original Article

Development and formative evaluation of a follow-up checklist for patients who have undergone sphincter-preserving surgery for rectal cancer based on the Supportive Care Framework

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ABSTRACT

Objective: Rectal cancer survivors who have undergone sphincter-preserving surgery (SPS) face complex challenges often under-addressed by surveillance-focused guidelines. Grounded in the Supportive Care Framework—which emphasizes meeting patients' holistic needs to improve outcomes—this study developed and evaluated a phase-specific follow-up checklist to standardize post-discharge management.
Methods: Using a mixed-methods design, an item pool was generated from a literature review ($n = 18$) and interviews ($n = 19$), mapped to the framework's four domains: physical, psychological, social, and informational. A two-round Delphi process ($n = 15$) refined the checklist. Consensus was reached through rigorous statistical validation of expert agreement. Finally, 12 patients and 7 nurses evaluated clinical acceptability and feasibility.
Results: Evidence synthesis highlighted gaps in physical function and psychosocial adaptation. Experts achieved high consensus, resulting in 52 items across four recovery phases. The checklist operationalizes tasks ranging from preventive stoma care and Low Anterior Resection Syndrome (LARS) management to social reintegration. Formative evaluation confirmed clinical utility and face validity, with an average nurse completion time of 22.7 minutes.
Conclusions: This theory-driven checklist complements surveillance guidelines by shifting focus to comprehensive survivorship care. It offers a structured workflow for the early detection of physical and psychosocial issues. Future research should prioritize multi-center psychometric validation and long-term effectiveness evaluation.

Introduction

Rectal cancer represents a significant global health burden and carries substantial long-term implications for patient quality of life.[1] Recent epidemiological trends indicate an increasing incidence in adults under 50, drawing growing public-health attention.[2,3] This demographic shift underscores the need to address long-term functional sequelae in a younger patient population. Rectal cancer constitutes about one-third of all colorectal cancers (CRC) and has an incidence of 13.9 per 100,000 in males and 8.6 in females.[4]

Its anatomical confinement within the bony pelvis necessitates a complex multidisciplinary treatment approach centered on radical surgical resection.[5] While sphincter-preserving surgery (SPS), now an option for up to 80% of patients, aims to preserve normal defecation pathways,[6] a gap often remains between functional preservation and postoperative quality of life.[7,8] Patients who have undergone radical resection for rectal cancer face severe complications, multidimensional burdens (physical, emotional, and financial), and substantial deterioration in quality of life.[9] Studies indicate that 5% to 10% of patients develop local recurrence postoperatively,[10] 60% to 80% develop varying degrees of bowel dysfunction,[7] and approximately 13% of colorectal cancer survivors present persistently low health-related quality of life (HR-QoL) or high psychological distress.[11]

Systematic follow-up is critical for early intervention, yet current systems often fail to support full-cycle postoperative management.[12–14] Protocols remain largely oriented to tumor markers and imaging, while needs specific to SPS survivors are frequently under-addressed. These unmet needs encompass bowel dysfunction (e.g., frequency, urgency,

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incontinence), urinary symptoms, sexual dysfunction, and body-image concerns, alongside requirements for post-treatment information, sustained multidisciplinary follow-up, and supportive services.[15,16] Clinical practice indicates that many patients do not receive systematic, dynamic assessment of bowel function, and existing pathways lack integrated psychosocial adaptation modules.[17] A framework driven exclusively by biomedical parameters may miss early signs of bowel dysfunction and psychological distress, delaying intervention and undermining confidence in recovery.[18]

An evidence-based follow-up checklist tailored to the SPS context could help bridge this gap by standardizing key tasks and facilitating timely identification of actionable issues.[19] To address workload concerns amid constrained oncology-nursing resources, the proposed checklist is designed to optimize rather than add burden, aiming to shift practice from routine data collection to targeted intervention so that limited resources can be focused on higher-risk patients. Evidence from other settings indicates that standardized checklists can improve the management of critical health care processes.[20–22] However, to our knowledge, no studies have developed or evaluated such a tool for rectal cancer after SPS. Moreover, existing follow-up checklists in other conditions often emphasize symptom monitoring while under-covering multidimensional needs.[5,23,24]

The Supportive Care Framework is a core component of clinical evidence-based nursing that broadly addresses patients' unmet needs throughout the continuum of care. It assists individuals and their families in coping with the cancer experience by meeting their physical, emotional, psychological, social, spiritual, and practical needs, as well as their informational needs.[25] It has shown advantages in chronic cancer care (e.g., breast and lung),[26,27] yet its postoperative application in rectal cancer remains unsystematic. In the SPS context, the Supportive Care Framework may offer distinct advantages for phase-adjusted care by addressing evolving needs from acute treatment toxicity to long-term rehabilitation.[28,29]

Building on this comprehensive framework, we developed a phase-specific checklist for SPS survivors by specifically operationalizing four domains—physical, psychological, social, and informational—to address the multidimensional challenges of SPS recovery using a two-round Delphi expert-consensus approach. This iterative, anonymous rating method employed structured feedback between rounds to build agreement based on a priori criteria for item retention. Subsequently, we conducted a formative evaluation of patient acceptability, nurse-perceived workflow feasibility, and time-on-task, aiming to support early detection of bowel dysfunction and systematic management of multidimensional needs.

Methods

Research design

We employed an exploratory sequential mixed-methods design, grounded in the Supportive Care Framework, to develop and preliminarily evaluate a follow-up checklist for rectal cancer patients after

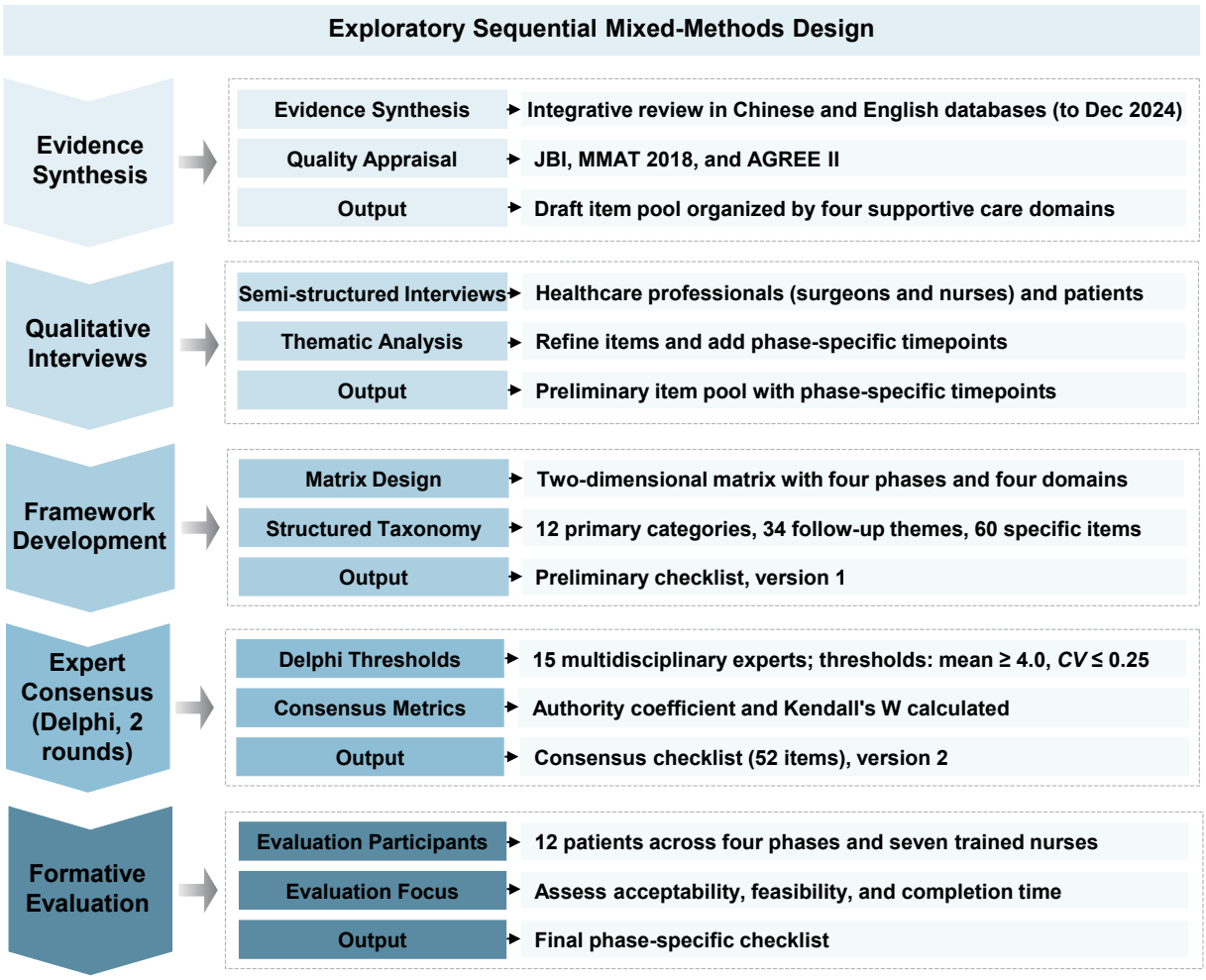

Fig. 1. Development pathway of the follow-up checklist using an exploratory sequential mixed-methods design. JBI, Joanna Briggs Institute; MMAT, Mixed Methods Appraisal Tool; AGREE II, Appraisal of Guidelines for Research and Evaluation II; CV, Coefficient of Variation; Kendall's W, Kendall's coefficient of concordance.

sphincter-preserving surgery.[30] A critical component of this design is "integration," which was achieved in this study through a "building" approach. This means that the findings from Phase 1 were explicitly used to inform the development of the quantitative instrument (the Delphi questionnaire) for Phase 2. Specifically, the integration process involved synthesizing the evidence from the integrative literature review with the themes and narratives derived from the qualitative interviews to generate the specific domains and items for the draft checklist. This multistage approach is consistent with guidance for developing structured clinical tools and complex interventions, where evidence generation, expert consensus, and early-phase field testing are integrated in sequence.[31,32] The process was organized into three main phases, comprising five sequential steps consistent with the mixed-methods integration pathway shown in Fig. 1: Phase 1 encompassed Steps 1–3: integrating findings from an integrative literature review and semi-structured interviews to complete framework development, where the preliminary checklist was built; Phase 2 corresponded to Step 4: establishing consensus through two rounds of Delphi expert consultation; Phase 3 involved Step 5: assessing feasibility via a formative evaluation.

Ethical considerations

This study received ethical approval from the Ethics Committee of the First Affiliated Hospital of Anhui Medical University (Approval No. PJ2025-01-21) and was performed in accordance with the Declaration of Helsinki. All participants were recruited from the Department of Gastrointestinal Surgery at the hospital and provided written informed consent after receiving a detailed explanation of the study's objectives, procedures, and voluntary nature. To ensure participant privacy, all personal data were anonymized using unique codes, stored securely on encrypted, password-protected computers accessible only to the core research team, and all findings are reported in an aggregated format to prevent individual identification.

Establishment of research team

A multidisciplinary team of eight was formed: one chief nurse specialist and one chief physician with extensive colorectal oncology experience, two nurse supervisors, and four postgraduate nursing researchers. The team's competencies aligned with the mixed-methods design. Senior members had $\geq$ 10 years of leading clinical studies (including qualitative projects) and oversaw overall study design and clinical content. Postgraduate researchers had completed graduate-level training in qualitative and quantitative research methods, including semi-structured interviewing, protocol development, survey instrument design, and Delphi procedures (questionnaire construction, structured feedback between rounds, and threshold-based item decision rules). All team members received study-specific training covering interview conduct, participant protection/ethics, and data handling and confidentiality. Nurse supervisors coordinated participant recruitment and site operations. This blended expertise supported methodological rigor across all phases—from evidence synthesis and stakeholder interviewing to expert consensus and formative evaluation.

Phase 1: Item pool generation

Integrative literature review

Search strategy. As the foundational phase for item generation, we conducted an integrative review following the framework established by Whittemore and Knafl, a methodology designed for synthesizing heterogeneous evidence across study types.[33] A comprehensive literature search was conducted across both Chinese and international databases, including CNKI, Wanfang Data, VIP, CBM, PubMed, MEDLINE, Embase, CINAHL, Web of Science, and the Cochrane Library. The search strategy combined Medical Subject Headings (MeSH) and free-text terms. For the Chinese databases, the search terms (translated from Chinese) included "rectal cancer", "rectal malignancy", "continuity of care", "follow-up", "return visit", "supportive care needs", "nursing needs", "unmet needs", and "needs assessment". For English databases, keywords included "rectal neoplasm", "rectal cancer", "rectum tumor", "follow-up", "continuity of care", "supportive care", "supportive care need*", "unmet need*", and "need* assessment*". Boolean operators (AND, OR) were used to combine the search terms appropriately. The search period ranged from the inception of each database to December 2024. The reference lists of the included studies were also manually screened to identify additional relevant publications. Given that our evidence synthesis was an integrative review to inform item generation—rather than a full systematic review or meta-analysis—we report the study-selection flow in Fig. 2 to enhance transparency.

Eligibility criteria. The literature inclusion criteria were as follows: (1) Thematic Focus: follow-up models, supportive care needs, continuity of care, or health management strategies for patients with rectal cancer after sphincter-preserving surgery; (2) Type of evidence: published Chinese and English guidelines, expert consensus, clinical decision-making, evidence summary, systematic review, randomized controlled experimental studies, and quasi-experimental studies; and (3) Publication in English or Chinese. The exclusion criteria were as follows: (1) Conference abstracts, letters, guideline interpretations, or non-peer-reviewed materials. (2) Studies rated as low quality. (3) Publications with insufficient information. (4) Literature for which the full text was not available.

Evidence extraction and assessment. Two researchers with training in evidence-based nursing independently screened the titles and abstracts, reviewed the full texts, and evaluated the study quality; a third reviewer resolved discrepancies. The extracted data included study characteristics, follow-up content, and supportive care needs. The methodological quality and risk of bias for primary studies were evaluated using the Joanna Briggs Institute (JBI) critical appraisal checklists specific to each study design, and the evidence levels were classified in accordance with the 2014 JBI Levels of Evidence.[34] For mixed-methods studies, we also applied the Mixed Methods Appraisal Tool (MMAT, 2018) to evaluate qualitative and quantitative components and their integration.[35] For clinical practice guidelines, quality appraisal used the Appraisal of Guidelines for Research and Evaluation II (AGREE II), which covers six domains—Scope and Purpose, Stakeholder Involvement, Rigor of Development, Clarity of Presentation, Applicability, and Editorial Independence.[36]

Because these instruments do not prescribe universal pass or fail cut-offs, we prespecified conservative operational thresholds a priori. This approach is consistent with methodological guidance for integrative reviews and aligns with the study's intended purpose of item generation rather than strict evidence grading.[33] The specific inclusion thresholds were: (1) JBI—studies with $\geq$ 2 items rated "No" or any single critical item rated "No" were classified as low quality and excluded; (2) MMAT—mixed-methods studies with $\geq$ 2 criteria rated "No/Can't tell" within any single component (qualitative, quantitative, or integration) were excluded (consistent with item-level judgments rather than a global score); (3) AGREE II—guidelines were excluded if Rigor of Development $<$ 30% (standardized 0% to 100%) or if the average of the six domain-standardized scores was $<$ 50%, or the overall recommendation was "not recommended." All extracted evidence was synthesized through team discussion and comparative analysis.

Semi-structured interviews

We used the Consolidated Criteria for Reporting Qualitative Research (COREQ) checklist to guide the reporting of our interview

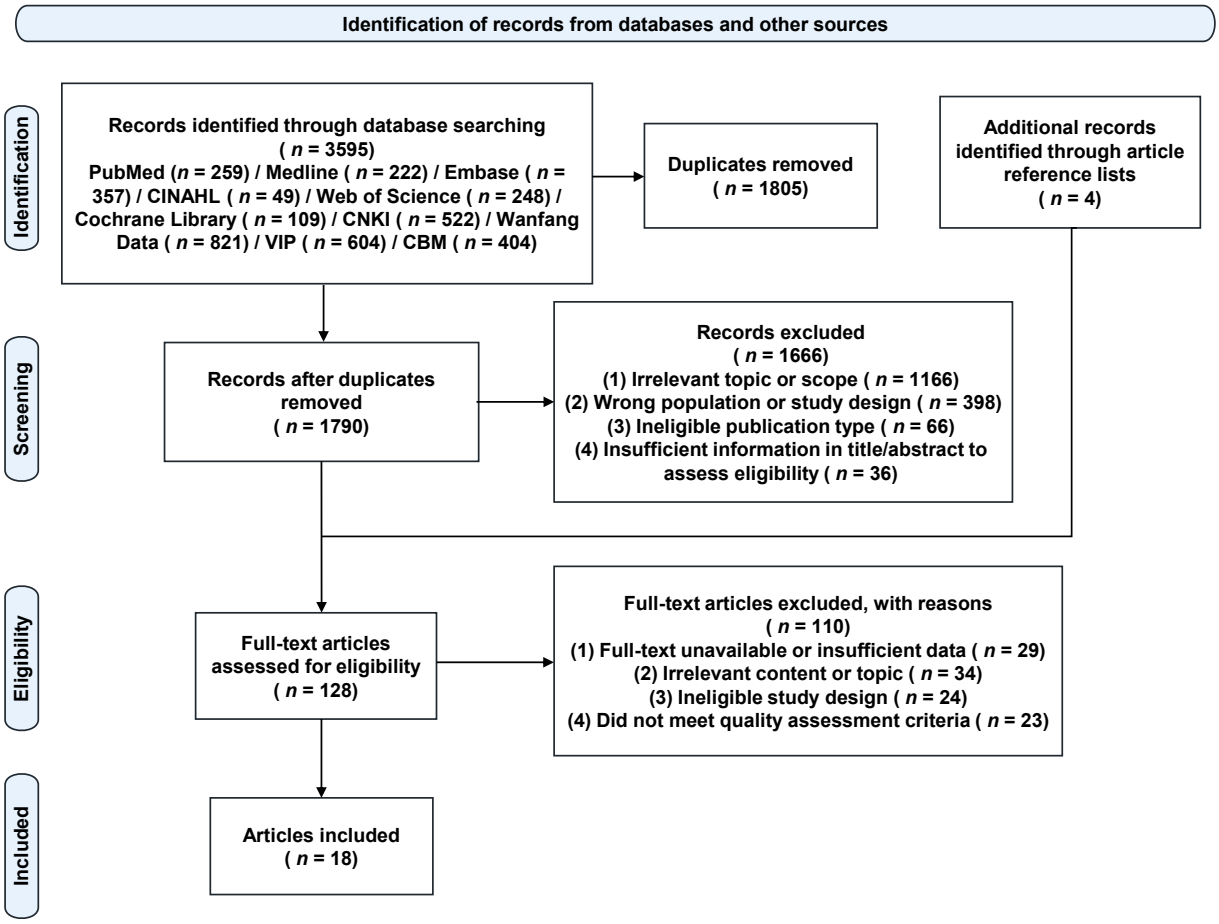

Fig. 2. Flow diagram of the literature selection process.

study.[37] To explore supportive care needs for checklist development, we conducted semi-structured interviews guided by four primary domains from the Supportive Care Framework, specifically physical, psychological, social, and informational, which were supplemented by an additional open-ended domain exploring systemic and comprehensive needs. The interview guide's validity was established through a multi-step, triangulated process involving a comprehensive literature review, adaptation from existing qualitative protocols in colorectal cancer survivorship studies,[38,39] review by an expert panel (two clinical nurse specialists, one colorectal surgeon, one qualitative researcher) for content validity, and pilot-testing with two patients and one healthcare professional to refine wording and flow.

Key topics explored included the lived experience of physical symptoms such as bowel dysfunction and the need for rehabilitation guidance; emotional and psychological distress following surgery; challenges in social participation and the support needs of caregivers; the adequacy, clarity, and preferred methods of information delivery; and perceived deficiencies in current follow-up services alongside priorities for designing future care programs. Appendices A and B provide a detailed mapping of interview items.

We used purposive sampling, aiming for maximum variation in demographics, to recruit a total of 19 participants, comprising 12 patients and 7 healthcare professionals, from a tertiary Grade A hospital in Anhui, China. Interviews, conducted from October to December 2024, lasted 30–40 minutes each. All participants provided written informed consent for audio recording.

Transcripts were analyzed using Braun and Clarke's six-phase inductive thematic analysis, a data-driven approach allowing themes to emerge directly from participant narratives.[40] To ensure rigor, the analysis followed a systematic procedure: first, two researchers independently coded the initial transcripts; second, they met to resolve discrepancies and finalize a coding framework; and third, one researcher subsequently applied this framework to all data, with regular consensus meetings with the second researcher to refine the thematic structure. Data were managed manually using Microsoft Word and Excel, and an audit trail with reflexive memos was maintained. Data saturation was deemed to have been reached after no new themes emerged in three consecutive interviews.[41] The resulting themes informed the preliminary item pool for the Delphi rounds.

Checklist framework development

To translate the evidence into a structured clinical tool, the research team developed a conceptual framework for the checklist based on a dual-dimensional matrix. The temporal dimension was designed to be phase-specific. The framework specified that the chronological stages would be initially framed by the follow-up schedules in major oncology guidelines,[15] and then populated and refined by the phase-specific supportive care needs identified in the evidence synthesis. The content dimension was derived from the four core domains of the Supportive Care Framework, encompassing the physical, psychological, social, and informational aspects. The framework further specified that an initial draft would be created by cross-tabulating these two dimensions to generate primary categories, populated with specific follow-up themes and items from the evidence synthesis, to be submitted for expert consultation.

Phase 2: Delphi expert consultation

Expert selection

The experts were purposively selected to ensure diverse professional perspectives and domain expertise. The inclusion criteria were as

follows: (1) experts with at least 10 years of experience in the diagnosis, treatment, or nursing care of rectal cancer and who held intermediate or senior professional titles; (2) physicians and psychological experts who were required to hold a master's degree or higher, while nurses held a minimum of a bachelor's degree; (3) psychological specialists who were certified Level II or III psychological counselors with $\geq$ 5 years of experience in psycho-oncology or psychological support for patients with chronic conditions; and (4) all the experts who provided written informed consent and voluntarily participated. These criteria ensured that the panel had both theoretical grounding and substantial practical insight into the supportive care of patients with rectal cancer.

Delphi questionnaire development

The Delphi expert consultation questionnaire consisted of three main sections: (1) a cover letter outlining the purpose, significance, and procedures of the study, emphasizing the need to develop a standardized follow-up checklist tailored to patients who have undergone sphincter-preserving rectal cancer surgery; (2) an item evaluation section containing a structured rating scale, where experts assessed the importance of each proposed follow-up indicator using a 5-point Likert scale (1 = "Very unimportant" to 5 = "Very important"). This section also included open-ended fields inviting qualitative feedback and suggestions for additions, deletions, or modifications to each item, thereby facilitating iterative refinement of the checklist; (3) an expert background form designed to collect detailed demographic and professional information (e.g., gender, age, academic degree, etc.), as well as two self-assessment components: (a) Judgment Basis—experts were asked to indicate the relative weight they assigned to various sources informing their responses (e.g., theoretical analysis, practical experience, subjective perception, and reference to domestic or international literature); (b) Familiarity Level—experts rated their level of familiarity with the topic on a 5-point scale (1 = "Very unfamiliar" to 5 = "Very familiar"). These elements were used to calculate the authority coefficient (*Cr*), which served as an indicator of panel credibility and expert input reliability.

Data collection

Two rounds of consultation were conducted between January and February 2025. The questionnaires were distributed via WeChat, email, or printed copies, depending on expert preference. After each round, the data were analyzed to assess central tendency (mean importance scores), dispersion (coefficient of variation), and consensus levels. Items with a mean score $\geq$ 4.0 and a coefficient of variation $\leq$ 0.25 were considered to have reached consensus. Expert suggestions and written comments were reviewed and discussed by the research team to inform revisions to the wording and structure of the items. After the second round, consensus was achieved for all indicators, and the final version of the follow-up checklist was established.

Phase 3: Formative evaluation

Participants

A stratified sampling strategy was employed to recruit 12 patients with rectal cancer who had undergone sphincter-preserving surgery and 7 registered nurses from the Department of Gastrointestinal Surgery. To ensure a rich diversity of feedback, patient recruitment ensured representation across four postoperative follow-up phases, ranging from 24 to 48 hours before discharge to more than 6 months post-discharge. The sample also included patients from three key surgical statuses, namely those with an unreversed stoma, those who had undergone stoma reversal, and those with no history of a stoma. Nurse participants were required to have at least 5 years of relevant clinical experience and completed study-specific training. All participants provided written informed consent.

The target sample sizes for this formative evaluation phase were determined based on the principle of information saturation, a standard approach for qualitative usability testing where the goal is to gather a rich diversity of feedback for iterative improvement rather than to achieve statistical generalization.[42] This qualitative-driven sample size is appropriate for the study's objective of identifying key usability issues and practical implementation considerations.

Evaluation instruments

As the primary objective of this phase was formative evaluation, two brief, investigator-developed instruments were used to gather targeted feedback on user experience: the Patient Acceptability Feedback Questionnaire and the Nurse-Perceived Workflow Feasibility Form.

The purpose of these tools was to collect preliminary, contextual insights to guide the final refinement of the checklist. In line with standard practices for the early-stage development of clinical tools, this formative phase prioritized the identification of practical implementation issues over resource-intensive psychometric validation.[43,44] Accordingly, the instruments were designed for an exploratory and pragmatic role.

The Patient Acceptability Feedback Questionnaire evaluated patient perceptions across five dimensions: content comprehensibility, communication experience, practicality, timeliness, and problem resolution. The Nurse-Perceived Workflow Feasibility Form captured nurse feedback on necessity, operational feasibility, time efficiency, logical coherence, and usefulness. Both used a 5-point Likert format to structure the feedback.

Procedures

Between March and April 2025, trained nurses implemented the draft follow-up checklist during routine postoperative care. Each patient participated in one follow-up session corresponding to their specific postoperative phase. Immediately following the session, patients completed the Patient Acceptability Feedback Questionnaire, and the administering nurse completed the Nurse-Perceived Workflow Feasibility Form. Additionally, both groups were asked open-ended questions to provide qualitative feedback on strengths, weaknesses, and suggestions for improvement, which were recorded by the research team for analysis. This qualitative feedback was subsequently analyzed using conventional content analysis.[45,46]

Statistical analysis

To ensure data quality, this study implemented dual-entry data management, a process in which data were entered independently by two researchers. The two resulting datasets were then compared to identify and resolve any discrepancies, thereby minimizing data entry errors. All data were managed and analyzed using Excel and SPSS version 27.0. Categorical data are presented as frequencies and percentages, whereas continuous variables are presented as the mean $\pm$ standard deviation. Expert demographic characteristics were summarized with frequency distributions and composition ratios. The response rate represented the expert engagement level. Expert authority was quantified through the authority coefficient (*Cr*),[47] which was calculated as $Cr = (Ca + Cs)/2$, where *Ca* denotes the judgment coefficient and *Cs* indicates the familiarity coefficient. Higher *Cr* values ($\geq$ 0.7 indicates acceptable reliability) reflect greater expert consensus reliability. The coordination of expert opinions was assessed via Kendall's coefficient of concordance (*Kendall's W*) and coefficient of variation (*CV*);[48,49] $P < 0.05$ was considered statistically significant.

Results

Phase 1: Item pool generation

Findings from the literature review

A total of 3595 records were identified through database searches (2351 in Chinese and 1244 in English). After deduplication, 1790

records were subjected to title and abstract screening. During the screening process, 1666 records were excluded, and 124 records were eligible for full-text review. Additionally, 4 relevant articles were included through reference tracing, resulting in 128 full-text reviews. Ultimately, 18 articles listed in Supplementary Table S1 were included in the final review, consisting of 16 original studies and 2 clinical practice guidelines (National Comprehensive Cancer Network [NCCN], European Society for Medical Oncology [ESMO]). Fig. 2 depicts the literature selection flow diagram. The most common reasons for full-text exclusion were irrelevant content or topic, full-text unavailable or insufficient data, ineligible study design, and not meeting quality-assessment criteria.

The integrative review, encompassing 16 peer-reviewed articles and 2 international clinical practice guidelines, was conducted to identify the key components of postoperative follow-up, including the core supportive care needs, for rectal cancer survivors after sphincter-preserving surgery. Guided by the principles of the Supportive Care Framework, the thematic analysis yielded five overarching themes: (1) Clinical Surveillance and Risk Management; (2) Symptom Management and Long-Term Sequelae; (3) Psychosocial and Emotional Support; (4) Information, Education, and Self-Management Empowerment; and (5) Care Coordination and Health System Support. A comprehensive summary of these themes is detailed in Supplementary Table S2.

Taken together, these themes reveal that the needs of survivors extend far beyond traditional oncological surveillance, demanding a holistic, patient-centered approach. Notably, several cross-cutting issues emerged as highly prominent across the literature: the management of Fear of Cancer Recurrence (FCR) as the primary psychosocial burden; the profound quality-of-life impact of Low Anterior Resection Syndrome (LARS) and other long-term sequelae; and a critical gap in health information and literacy, with patients consistently reporting a need for more reliable and individualized guidance. These findings underscore the necessity of moving beyond a purely biomedical model of follow-up to one that integrates comprehensive symptom management, psychological support, and robust patient education.

This comprehensive analysis provided the critical implications for our study: it underscored the necessity of developing a holistic, structured follow-up tool. To translate these analytical findings into a clinically practical framework for the checklist, the five themes were subsequently synthesized and reorganized into four user-centered dimensions: Physical, Psychological, Social, and Informational. This translational step, which connects the findings of the literature review to the structure of the developed checklist, is detailed in the checklist development section of our methodology.

Findings from the semi-structured interviews

Thematic analysis of interviews with a total of 19 participants (12 patients and 7 healthcare professionals) revealed five core domains of supportive care needs for patients following sphincter-preserving surgery for rectal cancer: Physical, Psychological, Social, Informational, and Systemic and Comprehensive Needs. Detailed demographic and clinical characteristics of the participants are presented in Supplementary Table S3 and S4. The data indicated that physical challenges, particularly those related to LARS, often served as a catalyst for needs across the other domains. A comprehensive overview of the identified domains, sub-themes, and representative quotations is provided in Supplementary Table S5.

Domain 1: Physical needs

Physical needs were consistently reported as the most immediate and distressing challenges. (1) Bowel dysfunction due to LARS: Most participants described this as a persistent loss of bodily control that disrupted daily life. As one patient described: “Sometimes I feel the urge to have a bowel movement but just can't; other times, it just leaks into my pants without me wanting it to” (P04). (2) Stoma-related complications: For patients with a temporary stoma, the period before reversal was characterized by anxiety related to self-care competence, with risks of leakage or prolapse identified as major threats to security. One participant recalled: “My stoma has prolapsed … as soon as I stand up and walk, that part of the intestine comes out” (P02).

Domain 2: Psychological needs

The physical challenges frequently translated into a psychological burden. (1) Need for emotional regulation support: Patients reported a struggle with anxiety and frustration, highlighting a need for support to manage symptom distress. As one patient noted: “When a person has to go to the toilet so frequently, of course they get frustrated and feel anxious” (P04). (2) Self-identity crisis: Particularly for younger or middle-aged patients, the altered body image (stoma) undermined their sense of self-worth and hope for the future. One participant expressed this loss of dignity: “Even though I know it's a temporary stoma, I still feel different from everyone else … it makes me feel a bit inferior” (P03).

Domain 3: Social needs

The impact extended to the social network. (1) Family support system: The home was identified as the pivotal sanctuary for physical care and emotional solace. A patient emphasized: “It's just me and my husband. The kids have to work … He is the one who takes care of me” (P02). (2) Social isolation: Conversely, the fear of losing bowel control acted as a barrier, causing patients to involuntarily withdraw from social engagements to avoid potential embarrassment. Another patient explained: “Because of the inconvenience with my bowels, I don't dare to go out, fearing I'll need a toilet midway” (P06).

Domain 4: Informational needs

Information was viewed as a tool for empowerment. (1) Specific and consistent guidance: Participants struggled with “information chaos” regarding diet and exercise, desiring clear, practical instructions. One patient stated: “The most important thing you can do is guide me on my diet … The information says this is forbidden … that isn't. It's completely confused me” (P03). (2) Timely communication channels: There was a critical need for professional bridges (e.g., WeChat groups) to cover the gap between hospital discharge and routine follow-up. A healthcare professional acknowledged: “Our routine follow-up calls have a time lag and cannot meet their immediate needs” (HP01).

Domain 5: Systemic and comprehensive needs

Finally, gaps within the health care system were highlighted. (1) Continuity of care: Patients expressed a need for better transition support from hospital to home. One patient highlighted this transition gap: “When I was discharged, nobody gave me any medication or told me when to come back … I had no idea what to do next” (P07). (2) Accessibility of technology: Concerns were raised about the cost and availability of rehabilitation tools like biofeedback. A professional suggested: “It would be better if there were more technologies and equipment … like the biofeedback therapy machine” (HP02). (3) Collaborative model: Participants called for consistent messaging across multidisciplinary teams (physicians and nurses) to ensure effective self-management. As one professional emphasized: “Subsequent health guidance is not just the work of nurses; doctors must also be involved … because sometimes what the doctor says is more persuasive” (HP02).

Development of the preliminary checklist

Representing the integration point of the mixed-methods design, the evidence synthesis from the literature review and interviews was used to build the preliminary checklist. We operationalized the qualitative themes into concrete checklist items. For instance, participant reports of “information chaos” regarding diet directly informed the generation of specific items for “Dietary guidance and recipes” in the Informational domain; similarly, patient descriptions of “losing bodily control” were translated into specific assessment items for LARS symptoms. This process confirmed a clear evolution of patient priorities across the recovery trajectory, leading to the finalization of four chronological stages: 24–48 hours before discharge, 1 week-3 months post-discharge, 3–6 months post-discharge, and more than 6 months post-discharge. By cross-tabulating the four content domains with the three post-discharge phases, a total of 12 primary categories were generated. These

categories follow a consistent pattern; for instance, the 'physical domain within the 1 week-3 months phase' and the 'Psychological domain within the 3–6 months phase'. Populating this matrix with the synthesized evidence resulted in the preliminary checklist comprising 34 follow-up themes and 60 specific items. This preliminary checklist was then advanced to the first round of the Delphi process.

Phase 2: Delphi expert consensus

Expert panel characteristics

The final consultation involved 15 experts from Anhui, Jiangsu, Zhejiang, and Shanghai. The expert demographics were as follows: age range, 35–57 years (mean 41.27 ± 6.93); professional experience, 10–37 years (mean 17.60 ± 8.02); the professional titles included 1 senior title, 10 associate senior titles, and 4 intermediate titles; the highest degrees included 6 doctoral, 5 master's, and 4 bachelor's degrees; and the research foci (multiple selections) included clinical nursing ($n = 7$), clinical medicine ($n = 6$), nursing research ($n = 2$), nursing education ($n = 2$), nursing management ($n = 2$), and psychology ($n = 2$). The detailed information can be found in Table 1.

Item refinement and consensus results

Expert positive coefficient. The study implemented a two-round Delphi consultation. In round 1, 17 questionnaires were distributed, of which 15 were returned and deemed valid, resulting in a valid response rate of 88.24%. In round 2, 15 questionnaires were distributed, with all 15 returned, resulting in a 100% valid response rate. The expert suggestion rate was 86.67% (13 experts) in Round 1 and 40.00% (6 experts) in Round 2.

Expert authority coefficient. For Round 1, the judgment coefficient (*Ca*) reached 0.947, the familiarity coefficient (*Cs*) was 0.913, and the authority coefficient (*Cr*) was 0.930. Round 2 showed improved coefficients: $Ca = 0.953$, $Cs = 0.940$, and $Cr = 0.947$.

Expert opinion coordination coefficient. The Kendall's coefficients of concordance for the two Delphi rounds were 0.152 and 0.201, respectively (both $P < 0.001$), indicating increasing concordance. The coefficients of variation ranged from 0 to 0.208 in Round 1 and 0.052 to 0.242 in Round 2.

Table 1
Characteristics of Delphi experts ($N = 15$).

Characteristics	*n*	%
Sex		
Male	7	46.67
Female	8	53.33
Age (years)		
35-39	8	53.33
40-49	5	33.33
50-57	2	13.33
Education background		
Bachelor	4	26.67
Master	5	33.33
Doctoral degree	6	40.00
Job title level		
Senior	1	6.67
Associate senior	10	66.67
Intermediate	4	26.67
Years of working experience		
10-19	12	80.00
20-29	1	6.67
30-37	2	13.33
Career focus field (multiple responses)		
Clinical medicine	6	40.00
Nursing practice	7	46.67
Nursing education	2	13.33
Nursing management	2	13.33
Nursing scientific research	2	13.33
Psychology	2	13.33

Percentages for "Career focus field" may exceed 100% due to multiple responses.

Revisions based on expert feedback. The two-round Delphi process generated a substantial volume of qualitative feedback (62 suggestions in Round 1 and 18 in Round 2), which guided the iterative refinement of the checklist. In the first round, key modifications involved deleting items deemed outside the core nursing scope (e.g., economic support), adding critical clinical assessments (e.g., nutritional status assessment), consolidating related items to reduce redundancy, and relocating items to more appropriate domains. The second round focused on further refinement, including the removal of several items for greater efficiency and the modification of wording for enhanced clinical precision. A detailed summary of all major revisions made during the Delphi process is provided in Supplementary Table S6.

Through two Delphi rounds, an expert consensus version of the follow-up checklist was established. This version solidified the hierarchical framework first conceptualized in the methods, comprising the 4 follow-up phases and the 12 primary categories derived from the matrix of phases and dimensions. Through expert feedback, the initial items were refined to a final set of 29 follow-up themes and 52 specific items. This expert-validated prototype was then advanced to the formative evaluation phase.

Phase 3: Formative evaluation

Following the initial development phases, a formative evaluation was conducted to assess the preliminary acceptability and feasibility of the draft checklist. This phase was designed to gather feedback from patients and nurses to inform the final refinement of the checklist.

Participant characteristics

The characteristics of the patients and nurses are summarized in Supplementary Table S7 and S8. Among the 12 patients (7 males, 5 females), the mean age was 60.5 ± 8.8 years (range 47–75). The majority of patients were married (91.7%), had at least a secondary-school education (83.3%), and resided in urban areas (58.3%). Surgical status was evenly distributed among unreversed preventive stoma ($n = 4$, 33.3%), reversed preventive stoma ($n = 4$, 33.3%), and no history of preventive stoma ($n = 4$, 33.3%). Patients were recruited across four postoperative follow-up phases: 24–48 hours before discharge ($n = 2$, 16.7%), 1 week–3 months post-discharge ($n = 5$, 41.7%), 3–6 months post-discharge ($n = 3$, 25.0%), and > 6 months post-discharge ($n = 2$, 16.7%).

All 7 participating nurses were female, aged 31–47 years (mean 38.1 ± 5.4), with a mean of 11.6 ± 4.9 years of clinical experience (range 6–20). The majority held the title of nurse-in-charge (71.4%), and all possessed a bachelor's degree or higher.

Quantitative findings on user acceptance

Quantitative feedback provided a general assessment of the checklist's initial reception. For patients ($n = 12$), the proportion of ratings of 'satisfied' or 'very satisfied' exceeded 80% across all five dimensions assessed, as detailed in Supplementary Table S9. For nurses ($n = 7$), clinical feasibility evaluations were similarly positive, as presented in Supplementary Table S10. Agreement rates (scores ≥ 4) exceeded 85.7% across all domains, with 'necessity' receiving the highest mean score (4.86 ± 0.38). The mean time recorded by nurses to complete the checklist was 22.7 ± 7.8 minutes. Overall, these descriptive statistics indicated a positive baseline user acceptance of the checklist's concept, providing a quantitative context for the qualitative feedback.

Qualitative feedback and final checklist revisions

The qualitative analysis identified specific directions for refinement. Feedback from patients indicated a need for the checklist's content to

Table 2
Finalized follow-up checklist for patients post-sphincter-preserving rectal cancer surgery based on the Supportive Care Framework.

Stage	Dimension	Follow-up theme	Specific Content	Mean ± SD	*CV*
24-48 hours before discharge			1 Confirm the patient's discharge plan through multidisciplinary team collaboration.	4.80 ± 0.41	0.086
			2 Inform the patient of follow-up details: content, methods, timing, and frequency.	4.80 ± 0.41	0.086
			3 Obtain the patient's contact information and documented consent for post-discharge follow-up.	4.47 ± 0.74	0.166
			Follow-up method: In-person; Frequency: Once.	4.60 ± 0.63	0.137
1 week - 3 months Post-discharge	Physical	Symptom assessment and management	1 Monitor defecation frequency, characteristics, regularity, and abnormalities; assess continence capacity and defecation difficulty severity.	4.60 ± 0.74	0.160
			2 Urinary Function: frequency, volume, color.	4.47 ± 0.52	0.115
			3 Incision healing Status: Assess for presence of redness, swelling, infection, exudate, etc.	4.93 ± 0.26	0.052
			4 Pain Management: Assess location/intensity, monitor changes, and adjust interventions dynamically.	4.93 ± 0.26	0.052
		Nutritional status assessment	5 Body Weight and changes: Instruct the patient to measure weight weekly at a fixed time using the same scale, record fluctuations, and provide personalized guidance.	4.67 ± 0.49	0.104
			6 Blood Biochemistry Monitoring:Regularly test albumin and prealbumin postoperatively. Provide personalized guidance based on quantitative results.	4.87 ± 0.35	0.072
			7 Dietary Intake: Encourage the patient to record daily food types and portions, and assess adequacy of fluid, protein, and other key nutrients.	4.60 ± 0.51	0.110
		Complications monitoring	8 Monitor postoperative complications, promptly identify symptoms and signs, and provide personalized management recommendations.	4.87 ± 0.52	0.106
		★Ostomy specialized care (sustained follow-up until stoma reversal.)	9 Monitor stoma status: height, diameter, mucosal color, and output characteristics (volume/color/consistency/odor).	4.93 ± 0.26	0.052
			10 Assess stoma function: frequency of accidental output leakage, nocturnal pouch-emptying frequency, and pouch ballooning frequency.	4.47 ± 0.64	0.143
			11 Screen for stoma complications and instruct patients on emergency management for specific conditions.	4.93 ± 0.26	0.052
			12 Manage peristomal skin conditions and provide patient guidance tailored to specific dermatologic issues.	4.80 ± 0.41	0.086
			13 Provide hands-on stoma care demonstration and daily appliance change guidance.	4.67 ± 0.62	0.132
			14 Assess patient/family stoma care competency and provide tailored reinforcement based on evaluation findings.	4.60 ± 0.63	0.137
	Psychological	Emotional health tracking	15 Dynamic tracking of anxiety/depression/insomnia with timely counseling or psychological referral upon symptom fluctuation.	4.60 ± 0.51	0.110
		Psychological expectation management	16 Inform patients about the postoperative symptom recovery timeline to mitigate psychological gaps and build recovery confidence.	4.60 ± 0.51	0.110
	Informational	Dietary guidance	17 Educate patients on mastering dietary principles covering food selection, eating methods, and dietary restrictions.	4.87 ± 0.35	0.072
		Bowel training guidance	18 ▲ Bowel reflex training: Use fixed daily time windows for toilet sitting (even without urge) to reinforce regularity.	4.27 ± 1.03	0.242
			19 Pelvic floor muscle training: Contract anal and perineal muscles (simulating a Kegel exercise), hold for 5 seconds → relax for 10 seconds; 10 repetitions per set, 3 sets daily.	4.73 ± 0.46	0.097
			20 ▲ Urge-control training: Use distraction techniques when urgency arises and delay defecation progressively to improve control.	4.73 ± 0.59	0.126
		Physical recovery guidance	21 Guide patients to gradually and progressively increase activity levels, establishing a light exercise regimen.	4.47 ± 0.64	0.143
			22 Alert patients to seek immediate medical attention if experiencing massive hematochezia, persistent high fever, or severe abdominal pain.	4.80 ± 0.56	0.117
	Social	Family support	23 Guide family members to master key postoperative recovery principles, provide nursing and emotional support techniques, and assist patients in rehabilitation exercises.	4.67 ± 0.49	0.104
			Follow-up modality: Telephone/video; Frequency: Once at 1 week, then monthly (1 week–3 months).	4.47 ± 0.74	0.166
3 - 6 months Post-discharge	Physical	Bowel function management	1 ▲ Monitor bowel function continuously.	4.47 ± 0.74	0.166
			2 ▲ Assess patient compliance with bowel training protocols and procedural correctness; rectify technical deviations.	4.47 ± 0.74	0.166
			3 ▲ Recommend biofeedback or electrical stimulation therapies based on individualized patient assessments.	4.27 ± 0.96	0.225

(*continued on next page*)

Table 2 (*continued*)

Stage	Dimension	Follow-up theme	Specific Content	Mean ± SD	*CV*
		Nutritional status assessment	4 Analyze patient's weight trend.	4.60 ± 0.51	0.110
			5 Monitor nutritional indicators based on risk stratification and nutrition-related clinical manifestations.	4.67 ± 0.62	0.132
			6 Assess the rationality of the patient's dietary structure	4.53 ± 0.52	0.114
		Sexual health management	7 Evaluate symptoms of sexual dysfunction, guide non-invasive interventions (e.g., pelvic floor muscle training, psychological counseling), and refer to urology if ineffective	4.20 ± 0.78	0.185
	Psychological	Body image adaptation	8 Assess patient's anxiety/shame regarding bodily changes, encourage self-acceptance, and guide progressive exposure exercises to overcome rejection	4.27 ± 0.80	0.187
	Informational	Dietary guidance	9 Provide individualized intensive guidance on dietary principles.	4.60 ± 0.63	0.137
			10 Instruct patients specifically on recording dietary reactions and educate on post-food intolerance response measures.	4.40 ± 0.83	0.188
		Medication management	11 Instruct on medication use, dosage adjustment, and side effect management.	4.73 ± 0.46	0.097
		Physical recovery guidance	12 Guide gradual transition to moderate-to-high intensity exercise (e.g., aerobic and strength training) based on individual recovery differences.	4.27 ± 0.80	0.187
		Recurrence warning	13 Educate patients on recognizing symptoms and signs of cancer recurrence and metastasis.	4.87 ± 0.35	0.072
		Follow-up Guidance	14 Encourage patient compliance with specialist reviews and strengthen pre-screening education.	4.67 ± 0.72	0.155
	Social	Home adaptation support	15 Optimize home environment to facilitate postoperative adaptation for patients and families.	4.27 ± 0.80	0.187
			Follow-up modality: Telephone + outpatient review; Frequency: Every 1.5 months (3–6 months).	4.20 ± 0.78	0.185
> 6 months Post-discharge	Physical	Bowel function	1 ▲ Evaluate bowel function management outcomes and promote patient adherence to maintenance training.	4.40 ± 0.83	0.188
		Nutritional monitoring	2 Conduct comprehensive nutritional status assessment (weight/metabolic/dietary logs) and develop individualized dietary recommendations.	4.60 ± 0.63	0.137
		Sexual health management	3 Re-evaluate sexual function recovery outcomes, sustain effective interventions, and refer complex cases to sexual health specialty.	4.13 ± 0.83	0.202
		Follow-up testing Guidance (Prescribed after physician evaluation)	4 Tumor markers (serum CEA, CA19-9).	4.80 ± 0.56	0.117
			5 Imaging studies (Contrast-enhanced CT of chest-abdomen-pelvis).	4.80 ± 0.56	0.117
			6 Colonoscopy.	4.80 ± 0.56	0.117
			7 Physical examination: Digital rectal examination, Abdominal palpation (performed by physician).	4.53 ± 0.74	0.164
	Psychological	Psychological resilience enhancement	8 Identify patients' hypervigilance to recurrence, cultivate mindfulness practice, and promote engagement in peer support groups.	4.20 ± 0.68	0.161
	Informational	Informational gap intervention	9 Assess information gaps in patients or families and provide timely targeted guidance.	4.27 ± 0.88	0.207
	Social	Role reintegration tracking	10 Monitor patients' reintegration into family functioning and reconstruction of social roles.	4.33 ± 0.90	0.208
		Social reintegration facilitation	11 Encourage patients to engage in community activities and support groups, rebuilding social interactions and sense of belonging	4.27 ± 0.80	0.187
			Follow-up modality: Telephone + outpatient review. Frequency: 6–12 months: every 3 months; Years 1–3: every 6 months; > 3 years: annually (≥ 5 years total).	4.60 ± 0.51	0.110

SD, standard deviation; CV, coefficient of variation; CEA, carcinoembryonic antigen; CA19-9, carbohydrate antigen 19-9. "▲" indicates that healthcare providers should ignore this item when following up with pre-stoma-reversal patients; "★" indicates stoma-specific items, and after stoma reversal surgery, restart follow-up from the 1 Week–3 Months Post-Discharge phase and omit all these items.

adapt to their evolving needs throughout the recovery process. Patients also noted the utility of the checklist's function in prompting psychosocial support discussions. Feedback from nurses centered on the need to balance the checklist's comprehensiveness with clinical efficiency, suggesting streamlining content by removing items redundant with the electronic health record.

These findings informed the final revisions to the checklist. Revisions included: adjusting phase-specific content to better align with the patient recovery trajectory, condensing several items to improve time efficiency, and augmenting psychosocial support prompts. This iterative refinement process culminated in the final version of the Post-Surgery Follow-up Checklist, which is detailed in Table 2. To further support implementation, a visual framework was also developed (Fig. 3), illustrating the recommended phased timeline, formats, and key focus of each phase.

Discussion

This study developed and preliminarily evaluated a phase-specific follow-up checklist for rectal cancer survivors after sphincter-preserving surgery (SPS). In contrast to previous tools that often lack theoretical grounding or specificity to the SPS population,[5,23,24] we employed a rigorous mixed-methods design to establish a clinical tool with strong content validity and initial face validity. Through the integration of qualitative patient insights and evidence-based literature foundations, this study ensured that the checklist content was not

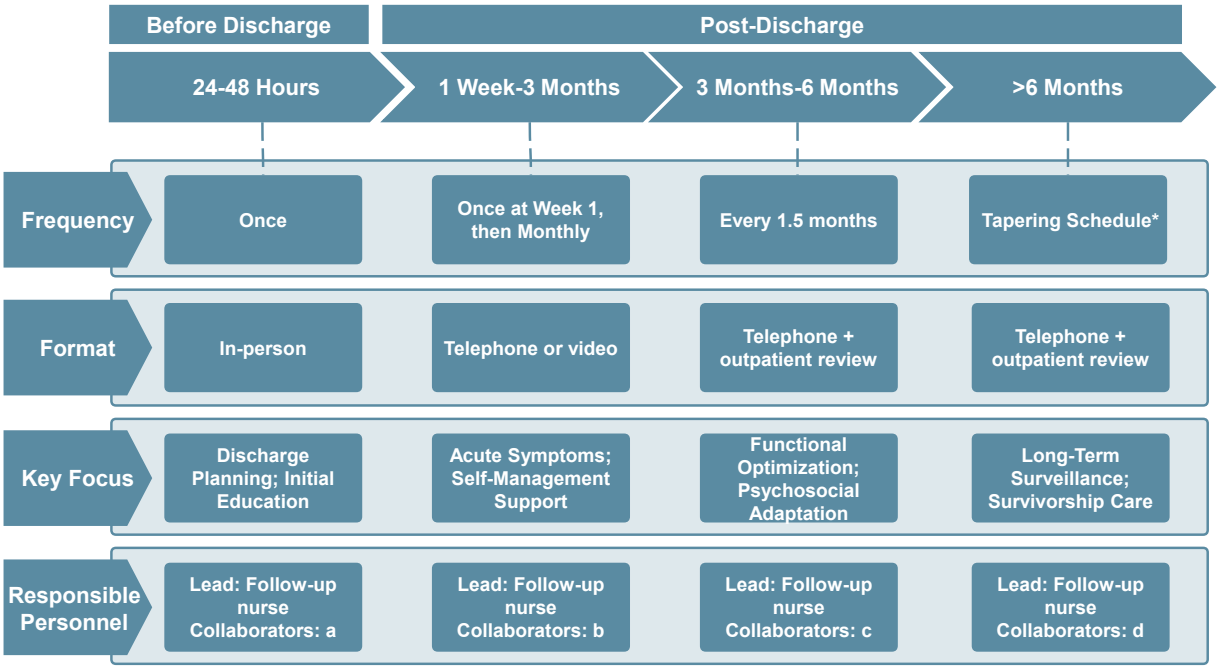

Fig. 3. Follow-up framework for rectal cancer after sphincter-preserving surgery. The responsible personnel row reflects implementation roles derived from the pilot usability evaluation and our routine clinical workflow; it was not part of the Delphi consensus output. Tapering Schedule*: 6–12 months: every 3 months; Years 1–3: every 6 months; > 3 years: annually (≥ 5 years total). Potential collaborators include: (a) surgeon, ward nurse, stoma therapy nurse (if applicable), dietitian; (b) surgeon, stoma therapy nurse (if applicable), dietitian, pelvic-floor physiotherapist, psycho-oncology professional; (c) surgeon, pelvic-floor physiotherapist, psycho-oncology professional, urologist/gynecologist (as needed); (d) surgeon/oncologist, gastroenterologist, primary-care physician, psycho-oncology professional, social worker.

merely theoretically derived but also grounded in the lived experiences of patients.

Main findings

Our main findings indicate that integrating patient-derived qualitative insights with evidence-based literature through a structured "building" strategy enabled the identification of phase-specific and context-sensitive survivorship needs. This "building" integration strategy allowed us to capture specific needs—such as the psychological burden of a temporary stoma—that might have been overlooked in a purely quantitative approach. Acknowledging that validity is intrinsic to usability, our findings demonstrate that establishing robust content validity through expert consensus serves as the fundamental prerequisite for a checklist's clinical utility.[50] The high consensus rates and positive user feedback (mean completion time 22.7 minutes) indicate that the checklist successfully translates evidence into a structured, implementable workflow. This study aligns with the "development phase" of the MRC framework for complex interventions,[32] providing the necessary structural foundation before proceeding to large-scale psychometric testing or effectiveness trials.

The application of the Supportive Care Framework served as the theoretical foundation for the checklist's design.[25] Drawing from this multidimensional model, we focused on the physical, psychological, social, and informational domains to structure our specific checklist items. While traditional surgical follow-up often prioritizes biomedical outcomes,[15,16] the adoption of this framework's four-domain structure ensured a holistic approach. Our interview results confirmed the framework's premise that patient needs are multidimensional and evolve over time, notably shifting from acute physical management (e.g., stoma care) in the early phases to complex psychosocial adaptation (e.g., social reintegration) in later phases.[28,29] Crucially, the theoretical focus on the "Informational" and "Social" domains allowed us to identify gaps often overlooked in routine care, such as the conflicting information regarding diet and the social isolation induced by bowel symptoms. By embedding these theoretical domains into the checklist, we ensure that non-physical needs, recognized as potent mediators of quality of life,[51,52] are systematically addressed.

A primary contribution of this checklist is its positioning as a structured complement to, rather than a replacement for, major oncology guidelines. Current guidelines, such as the NCCN Guidelines for Rectal Cancer and ESMO Clinical Practice Guidelines, provide detailed schedules for oncologic surveillance (e.g., CEA monitoring, CT scans) to detect recurrence.[1,4,53] However, they offer limited operational guidance on managing functional sequelae. For instance, while NCCN guidelines mention monitoring for "bowel dysfunction", they do not provide a specific protocol for assessing the nuances of LARS, such as clustering or urgency.[1,4] Our checklist bridges this gap by operationalizing these high-level recommendations into specific, actionable tasks. Where standard guidelines focus on "survival", this checklist focuses on "survivorship", ensuring that the goal of sphincter preservation is matched by the preservation of quality of life.

Implications for nursing practice and research

The checklist is designed to be used by healthcare professionals as a cognitive aid across the entire recovery trajectory, spanning from the pre-discharge phase to long-term follow-up. In the context of SPS, where 60% to 80% of patients experience LARS,[7] early identification is critical. Our qualitative findings highlighted that patients often underreport symptoms due to embarrassment. By explicitly including LARS-specific assessment items and linking them to phase-appropriate interventions (e.g., initiating pelvic floor muscle training in the early phase), the checklist establishes a standardized assessment logic for functional recovery. This targeted approach aims to shift practice from passive symptom reporting to proactive rehabilitation, potentially mitigating the long-term multidimensional burden described in the literature.[9]

Implementation feasibility is a key concern in resource-constrained settings. The pilot data suggests that the checklist is feasible for routine use, functioning to optimize rather than increase workload. By

standardizing the assessment logic, it allows healthcare professionals to focus limited resources on high-risk patients who trigger specific critical indicators (e.g., severe LARS score or high distress), rather than spending time on unstructured inquiries.[20–22] The manageable completion time supports its integration into nurse-led follow-up clinics or telehealth platforms, promoting continuity of care.

Limitations

Several limitations must be acknowledged. First, as aligned with the developmental and formative nature of this study, our primary objective was to establish content validity and assess the preliminary feasibility of the checklist. The feedback instruments used during the formative evaluation were designed specifically for this context and were not intended for full psychometric validation at this stage. Consequently, we did not conduct construct validity or discriminant validity testing, nor perform factor analytic procedures, which typically require a finalized tool and substantially larger samples. The descriptive statistics presented here (mean ± SD) therefore serve only as exploratory indicators to guide item refinement, rather than as evidence of underlying psychometric properties. Future large-scale validation work will incorporate comprehensive psychometric analyses, including construct validity, discriminant validity, internal consistency, and test–retest reliability. Additionally, participant responses may have been influenced by social desirability bias given the clinical setting. Consequently, we did not perform a randomized controlled trial to test effectiveness, which would require validated measures and a larger sample size. Second, the formative evaluation involved a small, single-center sample, which may limit the generalizability of the workflow feasibility findings to other hospital settings with different nurse-patient ratios. Third, the checklist's impact on long-term outcomes such as LARS severity or survival remains to be tested in future longitudinal studies. Finally, the tool currently relies on healthcare professional administration; future iterations could explore patient-reported digital integration to further reduce clinical workload.

Conclusions

This study addresses the critical gap in post-discharge management for rectal cancer survivors by developing a phase-specific, theory-driven follow-up checklist. By explicitly integrating qualitative patient insights, literature evidence, and quantitative expert consensus, this tool provides a structured complement to existing NCCN and ESMO surveillance guidelines. It facilitates the systematic assessment of multidimensional needs, specifically targeting the functional and psychosocial challenges unique to sphincter-preserving surgery that are often under-addressed in current practice. Future research should prioritize multi-center trials to evaluate the checklist's effectiveness on patient outcomes and conduct formal psychometric testing, paving the way for its integration into standard survivorship care pathways.

CRediT authorship contribution statement

Xinxin Huang: Conceptualization, Data Curation, Formal Analysis, Writing – Original Draft, Visualization. **Qingqiu Wu:** Methodology, Resources, Data Curation. **Min Hu:** Methodology, Validation. **Xinxin Zhang:** Methodology, Validation. **Beibei Song:** Resources, Investigation. **Qingqing Hong:** Resources, Investigation. **Shaohua Hu:** Project Administration, Supervision, Writing – Review & Editing. All authors have read and approved the final manuscript.

Ethics statement

This study was approved by the Ethics Committee of The First Affiliated Hospital of Anhui Medical University (Approval No. PJ2025-01-21) and was conducted in accordance with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. All participants provided written informed consent.

Data availability statement

The data that support the findings of this study are available from the corresponding author, SH, upon reasonable request.

Declaration of generative AI and AI-assisted technologies in the writing process

No AI tools/services were used during the preparation of this work.

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Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

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